

UML DIAGRAMS OVERVIEW FOR RESTAURANT APP MVP

1. Use Case Diagram

- Actors: Customer, Restaurant Manager.
- Use Cases: Browse Menu, Search Items, Filter Categories, View Item Details, Add to Cart, Apply Promo Code, Place Order, Reserve Table, Track Order Status, Update Order Status, Accept/Decline Reservations, Toggle Menu Availability.

2. Class Diagram

- Classes: User, Customer, Manager, Category, MenuItem, Cart, CartItem, Order, Reservation, Table.
- Description: Models relationships, attributes (with types like String, Number, Boolean), operations, and multiplicities between entities.

3. Sequence Diagram

- Scenario: Customer places an order.
- Flow: Customer taps Add to Cart -> Views Cart -> Applies Promo Code -> Chooses Dine-in/Takeaway -> Dispatches Order -> Receives Order Tracking Status.

4. State Machine Diagram

- Order Lifecycle States: Pending -> Preparing -> Ready -> Served -> (Cancelled).
- Transitions: Triggered by manager actions or automated timer intervals.

5. Component Diagram

- Architecture: React Native App -> App.js (Root / Navigators) -> Context Providers (AuthContext, ThemeContext, CartContext) -> Reducers & Custom Hooks -> Screens & Reusable Components (MenuItemCard).